

Problem E. Iva & Pav

Time Limit 5000 ms

Mem Limit 262144 kB

Iva and Pav are a famous Serbian competitive programming couple. In Serbia, they call Pav "papuca" and that's why he will make all of Iva's wishes come true.

Iva gave Pav an array a of n elements.

Let's define $f(l, r) = a_l \& a_{l+1} \& \dots \& a_r$ (here $\&$ denotes the [bitwise AND operation](#)).

Note that $f(l, r)$ is not defined when $l > r$.

Iva also gave Pav q queries.

Each query consists of 2 numbers, k and l , and she wants Pav to find the largest index r ($l \leq r \leq n$), such that $f(l, r) \geq k$.

Pav wants to solve this problem fast because he doesn't want to upset Iva. He needs your help.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of array a .

The second line of each test case contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — the elements of array a .

The third line of each test case contains a single integer q ($1 \leq q \leq 10^5$) — the number of queries Iva gave Pav.

The next q lines of each test case contains two numbers, l and k ($1 \leq l \leq n, 1 \leq k \leq 10^9$) — the left bound for the subsegment, and the integer k described in statement.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$. Also, it is guaranteed that the sum of q over all test cases does not exceed $2 \cdot 10^5$.

Output

For each query output maximal index r ($l \leq r \leq n$) such that $a_l \& a_{l+1} \& \dots \& a_r \geq k$.

If such r doesn't exist, output -1 .

Examples

Input	Output
3 5 15 14 17 42 34 3 1 7 2 15 4 5 5 7 5 3 1 7 4 1 7 5 7 2 3 2 2 7 19 20 15 12 21 7 11 4 1 15 4 4 7 12 5 7	2 -1 5 1 5 2 2 2 6 -1 5

Note

In the **first test case** $n = 5$, and the array $a = [15, 14, 17, 42, 34]$

The first query asks for the biggest index r such that the $f(1, r) \geq 7$.

$f(1, 1) = 15$, $f(1, 2) = 14$, $f(1, 3) = 0$ $f(1, 4) = 0$ $f(1, 5) = 0$, so $r = 2$ is the answer.

The second query asks for $f(2, r) \geq 15$. Since such r doesn't exist, the answer is -1 .

The third query asks for $f(4, r) \geq 5$. $f(4, 4) = 42$, $f(4, 5) = 34$, so $r = 5$ is the answer.

In the **second test case** $n = 5$, and the array $a = [7, 5, 3, 1, 7]$.

For the first query, $f(1, r) \geq 7$.

$f(1, 1) = 7$, $f(1, 2) = 5$, $f(1, 3) = 1$, $f(1, 4) = 1$, $f(1, 5) = 1$, so the answer to this query is 1.

For the second query, $f(5, r) \geq 7$.

$f(5, 5) = 7$, so the answer is 5.

For the third query, $f(2, r) \geq 3$.

$f(2, 2) = 5$, $f(2, 3) = 1$, $f(2, 4) = 1$, $f(2, 5) = 1$, so the answer is 2.